

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\Rightarrow f(x) = e^x - 2$$

$$\Rightarrow f(x_n) = e^{x_n} - 2 \quad \& \quad f'(x_n) = e^{x_n}$$

$$\Rightarrow x_{n+1} = x_n - 1 + 2e^{-x_n}$$